Evaluate the Final Model responses:

1. Whether the model appears to overfit

The model does not appear to overfit. Overfitting typically occurs when the training accuracy is significantly higher than the test accuracy, indicating that the model is memorizing the training data but not generalizing well. In this case, the training accuracy is approximately 68.6% while the test accuracy is slightly higher at 69.4%. Since the test performance is comparable to or slightly better than the training performance, the model appears to generalize reasonably well and does not show clear signs of overfitting.

2. Whether the test accuracy appears strong, weak, or unstable

The test accuracy of approximately 69.4% can be considered moderate but not strong. It is clearly above random guessing (which would be around 50% for a binary classification task), indicating that the model has learned some meaningful patterns. However, the accuracy is not high enough to be considered strong, especially for a predictive model that might be used in a real-world setting. The accuracy also appears stable, as it remains consistent across training epochs without large fluctuations.

3. Whether the dataset and feature set seem sufficient for a realistic admissions predictor

The dataset and feature set do not appear sufficient for building a realistic admissions predictor. While the model uses quantitative features such as GPA and GRE scores, it lacks many important factors that influence admissions decisions, such as research experience, letters of recommendation, statement of purpose, institutional reputation, and alignment with faculty interests. Additionally, the dataset appears to be imbalanced, with more rejected applicants than accepted ones, which can bias the model's predictions. As a result, the model is limited in its ability to capture the complexity of real admissions decisions.

Test the Model on Artificial Applicants response:

1. My findings:

The artificial applicant results show that the model does not always follow intuitive expectations based on academic strength alone. In this case, the stronger applicant with higher GPA and GRE scores was predicted as rejected, while the weaker applicant was predicted as accepted. This suggests that the model is influenced by other features such as degree type and citizenship, as well as patterns present in the dataset. For example, if the dataset contains more rejected international or PhD applicants, the model may learn to associate those characteristics with rejection. This highlights a limitation of the model, as it relies on correlations in the data rather than a true understanding of admissions criteria, and may produce counterintuitive or biased predictions.

Reflection responses:

1. What is useful about implementing a neural network manually?

Implementing a neural network manually is useful because it provides a deeper understanding of how the model actually works. By coding forward propagation, backpropagation, and gradient descent from scratch, I was able to see how inputs are transformed through layers, how errors are calculated, and how weights are updated to improve predictions. This level of transparency helps build intuition about how neural networks learn, which is often hidden when using high-level libraries like TensorFlow or PyTorch.

2. What are the limitations of using MSE for a binary classification task?

Mean Squared Error (MSE) is not ideal for binary classification because it treats the problem like a regression task rather than a probability-based classification problem. MSE does not strongly penalize incorrect confident predictions and can lead to slower or less effective learning. In contrast, loss functions like binary cross-entropy are specifically designed for classification and provide better gradients for updating model parameters, leading to improved performance.

3. What information is missing from this admissions dataset?

The dataset is missing many important factors that influence real graduate admissions decisions. These include research experience, quality of letters of recommendation, statement of purpose, undergraduate institution reputation, publications, internships, and alignment with faculty research interests. Without these features, the model only captures a small portion of what actually determines admissions outcomes, making it an incomplete representation of the real-world process.

4. Why might this model produce misleading results even if its test accuracy looks reasonable?

The model may produce misleading results because accuracy alone does not guarantee meaningful predictions. The dataset is imbalanced, with more rejected applicants than accepted ones, which can cause the model to favor predicting rejection. Additionally, the limited feature set means the model is making decisions based on incomplete information. As a result, even if the test accuracy appears reasonable, the model may not generalize well or reflect true admissions decisions, and its predictions could be biased or oversimplified.

5. What would you change if you wanted to make this model stronger or more realistic?

To improve the model, I would first replace MSE with a more appropriate loss function such as binary cross-entropy. I would also address class imbalance using techniques like resampling or class weighting. Adding more meaningful features, such as research experience and recommendation strength, would significantly improve realism. Additionally, I would experiment with more advanced models, regularization techniques, and validation methods to improve generalization and robustness.